International Institute of Medical Science & Technology Council

INTERNSHIP PROJECT REPORT

Data Science & Machine Learning (DSML)

Cohort 21 | Group 3

Project Title:

Forest Fire Ignition Risk Classification

SDG Goal: SDG 15: Life on Land

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Abstract

This project develops a machine learning based fire risk classification system using the Algerian Forest Fire Dataset, which contains 243 samples of daily meteorological readings and Fire Weather Index (FWI) components from two Algerian regions. Fire risk is categorised into three classes, Low, Moderate, and Severe, using established FWI threshold values. A Decision Tree and a Random Forest were trained and compared on this dataset. The Random Forest achieved 93.88 percent accuracy and an F1 score of 0.86 on the Severe class, outperforming the Decision Tree at 87.76 percent. FWI components, particularly ISI and FFMC, were identified as the strongest predictors of fire ignition risk. The Random Forest is recommended for deployment as a reliable fire risk early warning tool.

Chapter 1: Introduction

1.1 Background

Forest fires are one of the most destructive natural disasters, causing massive damage to biodiversity, soil health, air quality, and human settlements every year. Regions with dry summers and dense vegetation like the Algerian forests in North Africa are especially prone to large-scale fire events. Early detection and risk classification of such events is critical for disaster management agencies to respond in time.

Traditional fire monitoring methods rely on satellite imagery and manual field observation, which are often delayed or inaccurate. With the growing availability of meteorological data, machine learning models now offer a more practical and scalable approach to predicting fire ignition risk based on daily weather conditions. This project explores that direction by using a dataset of weather and FWI (Fire Weather Index) system parameters collected from two Algerian regions.

1.2 SDG 15 Life on Land

This project directly supports UN Sustainable Development Goal 15 Life on Land, which focuses on protecting, restoring, and promoting sustainable use of terrestrial ecosystems. Forest fires are one of the biggest threats to terrestrial biodiversity, and their increasing frequency due to climate change makes proactive risk management more important than ever.

By classifying daily meteorological conditions into low, moderate, or severe fire risk categories, this project aims to provide a data-driven tool that can assist forest management bodies in taking preventive action before a fire event occurs. This aligns with SDG 15's targets around combating desertification and biodiversity loss caused by land degradation and uncontrolled fires.

1.3 Problem Statement

The core problem this project addresses is: given a set of daily meteorological measurements, can we accurately classify the fire ignition risk for that day as Low, Moderate, or Severe? Existing binary approaches (fire vs. no fire) are too coarse for operational use emergency teams need to know not just whether risk exists, but how serious it is. A three-class system gives much more actionable information.

Additionally, Severe risk days are rare compared to Low risk days, which creates a class imbalance problem. Misclassifying a Severe day as Moderate can have real consequences, so the model needs to be specifically evaluated and tuned for high-risk detection performance, not just overall accuracy. Categorical cross entropy loss and per-class F1 scores are used to measure this carefully.

1.4 Objectives

The main objectives of this project are as follows:

1. To classify daily meteorological conditions into three fire risk categories: Low, Moderate, and Severe.
2. To evaluate model performance using categorical cross entropy loss and per-class precision, recall, and F1-score.
3. To engineer a Temperature-Humidity interaction feature and assess its contribution to improving Severe class detection.
4. To compare Decision Tree and Random Forest classifiers and identify which is better suited for deployment in real-world fire monitoring scenarios.
5. To analyze feature importance and understand which meteorological variables drive fire risk most significantly.

1.5 Scope

This project is scoped to the Algerian Forest Fire dataset which covers two regions, Bejaia and Sidi Bel-Abbes, over the summer months of 2012. The analysis is limited to tabular meteorological data and does not include satellite imagery or geospatial data. The models built are supervised classifiers (Decision Tree and Random Forest) and the focus is on classification accuracy, particularly for the Severe risk class.

The project does not attempt real-time prediction or deployment as a live system. It serves as a proof-of-concept internship project demonstrating that weather data alone can be used to build a reasonably accurate fire risk classifier, with potential to be extended further with more data and advanced models.

Chapter 2: Literature Review

2.1 Existing Work and Research

Several studies have explored the use of the Algerian Forest Fire dataset for fire prediction tasks. Abid (2019) introduced the dataset and demonstrated its usefulness for binary classification using simple regression and decision tree models, achieving around 80–85% accuracy. The work highlighted that FWI system components like ISI and FFMC are strongly correlated with fire occurrence, which was consistent with field observations.

Later work by Faroudja and Issam (2019) used a combination of SVM and KNN classifiers on the same dataset and pushed accuracy higher, around 98% for binary classification. However, binary classification (fire/no fire) is limited in practical terms since it doesn't tell responders how severe the risk is. More recent papers have started moving toward multiclass approaches but these remain relatively underexplored compared to binary formulations.

2.2 Machine Learning in Fire Risk Prediction

Machine learning has been applied to wildfire risk prediction in various forms across different regions. Random Forest is consistently one of the top performers in forest fire-related classification tasks due to its ensemble nature, which handles noisy meteorological data well. Studies from Portugal and California have shown that RF models outperform single decision trees, especially when dealing with imbalanced class distributions where fire events are rare.

Gradient boosting methods like XGBoost and LightGBM have also shown strong results but require more careful hyperparameter tuning and can be harder to interpret. For this project, Decision Tree was chosen as a baseline for its interpretability, while Random Forest was selected as the primary model. Both are well-suited to tabular meteorological data without needing extensive preprocessing compared to neural networks.

2.3 Multiclass Classification with Cross Entropy

Multiclass classification involves predicting one of three or more categories for each input. In this project, the three classes are Low, Moderate, and Severe fire risk. A common loss function used to evaluate such models is categorical cross entropy, which measures how well predicted probability distributions match the actual class labels. Lower cross entropy indicates the model is more confident and correct in its predictions.

For tree-based models like Decision Tree and Random Forest, scikit-learn internally uses Gini impurity or entropy during training, but categorical cross entropy can be computed post-training using the predicted class probabilities output from predict_proba(). This allows a fair comparison of model calibration alongside standard accuracy and F1 metrics, especially useful when the Severe class is a minority and needs to be weighted appropriately.

2.4 Research Gap

Most existing work on the Algerian Forest Fire dataset is focused on binary classification and does not address the severity grading of fire risk. The Severe class, which is the most operationally important, is often underrepresented in model evaluations. Very few papers specifically analyze the effect of engineered features like Temperature-Humidity interaction terms on Severe class detection performance.

This project attempts to fill that gap by framing fire risk as a three-class problem, evaluating with per-class metrics, and systematically studying whether adding interaction features improves detection of the Severe class. The comparative study between Decision Tree and Random Forest also adds practical value by identifying which model is more appropriate for deployment versus explanation purposes.

Chapter 3: Dataset Description and Exploratory Data Analysis

3.1 Dataset Overview

The dataset used in this project is the Algerian Forest Fire dataset, which contains meteorological and Fire Weather Index (FWI) system measurements recorded daily over the summer of 2012. The final processed file used for modeling is forest_fire_scaled.csv, which contains 243 samples and 20 columns after preprocessing and feature engineering.

The data covers four months, June, July, August, and September, with roughly equal representation across months (around 60–62 records per month). All numeric features have been standardized using StandardScaler, so the values in the dataset are z-scores rather than raw measurements. The original meteorological variables include Temperature, Relative Humidity (RH), Wind Speed (Ws), Rain, and FWI system components: FFMC, DMC, DC, ISI, BUI, and FWI.

3.1.1 Column Description

Column	Type	Description
day, month, year	Temporal	Date of observation. Year is 2012. Months cover June to September (6–9).
Temperature	Meteorological	Daily maximum temperature (scaled). Higher values indicate hotter days with increased fire risk.
RH	Meteorological	Relative Humidity (%). Low humidity dries out vegetation making it more flammable.
Ws	Meteorological	Wind speed (km/h, scaled). Higher wind speeds accelerate fire spread.
Rain	Meteorological	Daily rainfall (mm, scaled). Presence of rain generally suppresses fire risk.
FFMC	FWI Component	Fine Fuel Moisture Code: moisture content of surface litter. Highly correlated with ignition likelihood.
DMC	FWI Component	Duff Moisture Code: moisture in loosely compacted organic layers below surface.

DC	FWI Component	Drought Code: moisture in deep organic layers. Reflects long-term drought effects.
ISI	FWI Component	Initial Spread Index: expected rate of fire spread. Top feature in RF importance (25.76%).
BUI	FWI Component	Buildup Index: total amount of fuel available for combustion.
FWI	FWI Component	Fire Weather Index: overall fire intensity rating. Used as the basis for deriving the Fire_Risk target variable.
Temp_Humidity	Engineered	Temperature × Humidity interaction term. Captures combined thermal stress effect.
Temp_Humidity_Ratio	Engineered	Ratio of Temperature to Humidity. High values indicate hot and dry conditions.
Heat_Index	Engineered	Derived heat index combining temperature and humidity effects on perceived heat stress.
Fire_Risk_Label	Target	Fire risk class label: Low, Moderate, or Severe. Derived from FWI thresholds.
Fire_Risk_Encoded	Target	Numeric encoding used for model training: Low=0, Moderate=1, Severe=2.

3.2 Target Variable Distribution

The target variable Fire_Risk_Encoded has three classes derived from FWI thresholds. The class distribution across 243 samples is as follows:

Class	Encoded	Count	Percentage
Low	0	104	42.8% (majority class)
Moderate	1	90	37.0%
Severe	2	49	20.2% (minority class)

The class distribution shows a mild imbalance. Low risk days are the most common (42.8%), followed by Moderate (37.0%), and Severe being the least frequent at only 20.2%. This imbalance is important because the Severe class is operationally the most critical. A missed Severe day can have serious real-world consequences. Stratified splitting was used during train/test division to ensure all three classes were proportionally represented in both sets.

3.3 Key EDA Findings

During exploratory data analysis, several patterns were identified that helped guide the modeling decisions:

1. FWI components are the strongest predictors: ISI and FFMC showed the highest correlation with the Fire_Risk target. This was later confirmed by Random Forest feature importance where ISI (25.76%) and FFMC (24.33%) together accounted for over 50% of predictive power.
2. Even monthly distribution: The dataset has approximately equal records per month (60–62 per month across June, July, August, and September), meaning there is no temporal bias from one month dominating the dataset.
3. Temperature and Humidity interaction: High temperature combined with low humidity consistently appeared in Severe risk days. This motivated the creation of the engineered features Temp_Humidity, Temp_Humidity_Ratio, and Heat_Index to explicitly capture this relationship.
4. Rain is mostly zero: The Rain column is heavily zero-inflated across the summer months. On the few days where rainfall was recorded, fire risk was predominantly Low, confirming its suppressive effect on fire ignition.
5. Severe class overlap with Moderate: EDA showed that the boundary between Moderate and Severe risk is not always clearly separable using simple meteorological variables alone. This is why FWI-derived components are critical as they encode the cumulative dryness and fire behavior in ways that raw temperature or humidity cannot.

Overall, the EDA confirmed that the dataset is suitable for multiclass classification, that FWI system features should be retained as primary predictors, and that engineered temperature-humidity features are worth including to improve Severe class detection.

Chapter 4: Data Preprocessing and Feature Engineering

4.1 Data Cleaning

The raw dataset had several issues that needed to be fixed before modeling. Column names had trailing whitespace, some rows had the string “Classes” as a value (header row repeated in data), and a few rows had null values. All of these were cleaned systematically. Additionally, all non-target columns were explicitly cast to numeric type to avoid dtype-related issues during model training.

Code:

```
import pandas as pd
import numpy as np

df = pd.read_csv("algerian_forest_fire_with_features.csv")

# Strip whitespace from column names
df.columns = df.columns.str.strip()

# Remove repeated header rows in data
df = df[df['Classes'] != 'Classes']

# Drop null rows
df = df.dropna()

# Convert all non-target columns to numeric
for column in df.columns:
    if column != 'Classes':
        df[column] = pd.to_numeric(df[column])
```

4.2 Label Encoding

The original Classes column had string values “fire” and “not fire”. These were mapped to binary integers (1 and 0) for compatibility with sklearn models. Separately, the Fire_Risk column (Low / Moderate / Severe) was encoded using LabelEncoder to produce Fire_Risk_Encoded (0, 1, 2), which was used as the final target variable for multiclass classification.

Code:

```
# Map fire/not fire to binary
df['Classes'] = df['Classes'].map({'not fire': 0, 'fire': 1})


# Encode Fire_Risk (Low/Moderate/Severe) using LabelEncoder
from sklearn.preprocessing import LabelEncoder


le = LabelEncoder()
df["Fire_Risk_Encoded"] = le.fit_transform(df["Fire_Risk"])


print("Label Encoding mapping:")
for cls, val in zip(le.classes_, range(len(le.classes_))):
    print(f"  {cls} -> {val}")
```

4.3 **Feature Engineering: Temperature Humidity Interaction**

Three new features were engineered to explicitly capture the relationship between temperature and humidity, which EDA had shown was particularly relevant for Severe risk days. These are: Temp_Humidity (product), Temp_Humidity_Ratio (ratio), and Heat_Index (a standard meteorological formula). All three were appended to the dataframe before scaling.

Code:

```
# Temperature x Humidity product
df['Temp_Humidity'] = df['Temperature'] * df['RH']


# Temperature / Humidity ratio
df['Temp_Humidity_Ratio'] = df['Temperature'] / df['RH']


# Heat Index formula
df['Heat_Index'] = 0.5 * (df['Temperature'] + 61.0 + ((df['Temperature']-
68.0)*1.2) + (df['RH']*0.094))
```

4.4 Feature Scaling

StandardScaler was applied to all numeric feature columns to normalize them to mean=0 and std=1. This ensures that features with large ranges (like DC or BUI) do not dominate the model over smaller-range features. The scaled dataframe was saved as forest_fire_scaled.csv for use in model training. MinMaxScaler was also tested for comparison but StandardScaler was chosen as the final approach.

Code:

```
from sklearn.preprocessing import StandardScaler


cols_to_scale = ["Temperature", "RH", "Ws", "Rain", "FFMC", "DMC", "DC",
                 "ISI", "BUI", "FWI", "Temp_Humidity", "Temp_Humidity_Ratio",
"Heat_Index"]


scaler = StandardScaler()
df_scaled = df.copy()
df_scaled[cols_to_scale] = scaler.fit_transform(df[cols_to_scale])
df_scaled.to_csv("forest_fire_scaled.csv", index=False)
```

4.5 Train Test Split

The scaled dataset was split into training (80%) and testing (20%) sets using train_test_split with stratify=y to ensure proportional class representation in both splits. This is important given the mild class imbalance. Without stratification, the Severe class (only 49 samples) could end up underrepresented in the test set, making evaluation unreliable. A fixed random_state=42 was used for reproducibility.

Code:

```
from sklearn.model_selection import train_test_split


X = df_scaled.drop(columns=["Fire_Risk", "Fire_Risk_Encoded",
"Fire_Risk_Label"])
y = df_scaled["Fire_Risk_Encoded"]


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)



print("Train size:", X_train.shape, "| Test size:", X_test.shape)
```

Chapter 5: Model Building and Training

5.1 Model Selection

For this project, two supervised classification models were selected to predict forest fire ignition risk, specifically the Decision Tree Classifier and the Random Forest Classifier. Both models were chosen based on their suitability for multi-class classification tasks and their compatibility with the structured, tabular nature of the Algerian Forest Fire Database

Decision Tree was selected as the baseline model due to its simplicity and interpretability. It builds a tree-like structure of if-else rules based on feature thresholds, making it easy to understand which conditions lead to each fire risk level. It requires no feature scaling and handles both numeric and categorical data well.

Random Forest was selected as the final model because it addresses the key weakness of Decision Trees, which is overfitting. By training an ensemble of 100 decision trees on random subsets of the data and averaging their predictions, Random Forest produces a more stable and accurate model. It also provides feature importance scores, which help identify the most significant meteorological variables driving fire risk.

Both models were evaluated on the same train/test split to allow a direct and fair comparison of their performance metrics.

5.2 Model Architecture

5.2.1 Decision Tree Architecture

The Decision Tree model was configured with the following hyperparameters:

```
DecisionTreeClassifier(
    criterion  = "gini",    # Gini impurity for splitting
    max_depth  = 5,         # Limits tree depth to prevent overfitting
    random_state = 42       # Fixed seed for reproducibility
)
```

The Gini criterion measures the impurity of a node. The model splits on the feature that produces the purest child nodes. A max_depth of 5 was chosen to control model complexity after observing that deeper trees began to overfit on the training data.

5.2.2 Random Forest Architecture

The Random Forest model was configured with the following hyperparameters:

```
RandomForestClassifier(
    n_estimators = 100,    # Number of decision trees in the ensemble
    criterion    = "gini", # Gini impurity for each tree
    max_depth    = None,   # Trees grow fully unless limited by min samples
    random_state = 42      # Fixed seed for reproducibility
)
```

With n_estimators=100, the model builds 100 independent trees, each trained on a bootstrapped sample of the training data with a random subset of features considered at each split. The final prediction is determined by majority vote across all trees, which significantly reduces variance and improves generalization.

5.3 Training Process

5.3.1 Dataset Split

The dataset was split into training (80%) and testing (20%) sets using stratified sampling to ensure that all three fire risk classes, Low, Moderate, and Severe, were proportionally represented in both splits. A fixed random_state=42 was used for reproducibility.

```
from sklearn.model_selection import train_test_split

X = df.drop(columns=["Fire_Risk", "Fire_Risk_Label",
                      "Fire_Risk_Encoded", "Classes",
                      "FWI", "day", "month", "year"])
y = df["Fire_Risk_Label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
print("Train size:", X_train.shape, "| Test size:", X_test.shape)
```

5.3.2 Training the Decision Tree

The Decision Tree model was fit on the training data as follows:

```
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(
    criterion="gini", max_depth=5, random_state=42
)
dt_model.fit(X_train, y_train)
print("Decision Tree training complete.")
```

5.3.3 Training the Random Forest

The Random Forest model was fit on the same training data:

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=100, criterion="gini",
    max_depth=None, random_state=42
)
rf_model.fit(X_train, y_train)
print("Random Forest training complete.")
```

Both models were trained on identical splits, ensuring that any difference in performance metrics during evaluation is purely due to the model architecture and not variations in training data.

5.4 Decision Tree Model

The Decision Tree classifier was trained on the Algerian Forest Fire Dataset to predict fire ignition risk across three categories: Low, Moderate, and Severe. The model was built using scikit-learn's DecisionTreeClassifier with a Gini impurity criterion and a maximum depth of 5 to prevent overfitting.

5.4.1 Importing Libraries

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix,
accuracy_score
```

5.4.2 Preparing Features and Target

The target variable is Fire_Risk_Label (0 = Low, 1 = Moderate, 2 = Severe). Columns not useful for prediction are dropped.

```
X = df.drop(columns=["Fire_Risk", "Fire_Risk_Label", "Fire_Risk_Encoded",
                      "Classes", "FWI", "day", "month", "year"])
y = df["Fire_Risk_Label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

5.4.3 Model Training

```
dt_model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)
dt_model.fit(X_train, y_train)
```

5.4.4 Model Evaluation

```
y_pred = dt_model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

The Decision Tree achieved 85.71% accuracy on the test data. The confusion matrix showed that the model correctly classified most Low and Moderate risk samples, but misclassified 3 Severe risk days as Moderate, which is a critical error in a real fire prediction scenario.

5.4.5 Sample Prediction

```
sample_data = pd.DataFrame({
    "Temperature": [22, 32, 42],
    "RH":          [75, 40, 15],
    "Ws":          [2,  4,  8],
    "Rain":        [2,  0,  0],
    "FFMC":        [65, 85, 96],
    "DMC":         [80, 120, 200],
    "DC":          [150, 350, 600],
    "ISI":         [1,  5,  12],
    "BUI":         [70, 140, 250],
    "Temp_Humidity":       [22*75, 32*40, 42*15],
    "Temp_Humidity_Ratio": [22/75, 32/40, 42/15],
    "Heat_Index": [
        0.5*(22+61+((22-68)*1.2)+(75*0.094)),
        0.5*(32+61+((32-68)*1.2)+(40*0.094)),
        0.5*(42+61+((42-68)*1.2)+(15*0.094))
    ]
})
predictions = dt_model.predict(sample_data)
label_map = {0: "Low", 1: "Moderate", 2: "Severe"}
for i, p in enumerate(predictions):
    print(f"Sample {i+1} Risk Level:", label_map[p])
```

5.5 Random Forest Model

The Random Forest classifier is an ensemble learning method that builds multiple decision trees during training and combines their outputs by majority voting. It was used as the final model in this project due to its superior accuracy and reduced overfitting compared to a single Decision Tree.

5.5.1 Importing Libraries

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix,
accuracy_score
```

5.5.2 Preparing Features and Target

The same feature set and train/test split used for the Decision Tree is applied here to ensure a fair comparison.

```
X = df.drop(columns=["Fire_Risk", "Fire_Risk_Label", "Fire_Risk_Encoded",
                      "Classes", "FWI", "day", "month", "year"])
y = df["Fire_Risk_Label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

5.5.3 Model Training

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    criterion="gini",
    max_depth=None,
    random_state=42
)
rf_model.fit(X_train, y_train)
```

5.5.4 Model Evaluation

```
y_pred_rf = rf_model.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))
print("Classification Report:")
print(classification_report(y_test, y_pred_rf))
```

The Random Forest model outperformed the Decision Tree across all evaluation metrics. It achieved higher overall accuracy, improved precision and recall for Severe risk cases, and provided feature importance scores to identify the most influential meteorological variables.

5.5.5 Feature Importance

```
import pandas as pd
import matplotlib.pyplot as plt


feature_importances = pd.Series(
    rf_model.feature_importances_, index=X.columns
).sort_values(ascending=False)


print("Feature Importances:")
print(feature_importances)


feature_importances.plot(kind="bar", title="Feature Importance - Random
Forest")
plt.tight_layout()
plt.show()
```

5.5.6 Sample Prediction

```
predictions_rf = rf_model.predict(sample_data)
label_map = {0: "Low", 1: "Moderate", 2: "Severe"}
for i, p in enumerate(predictions_rf):
    print(f"Sample {i+1} Risk Level:", label_map[p])
```

Chapter 6: Results and Evaluation

6.1 Evaluation Overview

This chapter presents the evaluation results for the Random Forest (RF) classifier, which was selected as the final model for forest fire ignition risk prediction. The Random Forest outperformed the Decision Tree across all key metrics, making it the recommended model for deployment. Evaluation was performed on the Algerian Forest Fire Dataset with a stratified 80/20 train/test split.

6.2 Dataset Summary

Property	Value
Total Samples	243
Features Used	15 (DT) / 12 (RF)
Target Classes	Low (104), Moderate (90), Severe (49)
Train/Test Split	80% / 20% (stratified)
Scaling Applied	StandardScaler
Class Imbalance	Mild, Severe is minority class (20%)

6.3 Overall Performance Metrics

The table below compares both models across key evaluation metrics. Random Forest wins on every metric, confirming its superiority for this task.

Metric	Decision Tree	Random Forest	Winner
Accuracy	87.76%	93.88%	RF ✓
Precision	87.76%	94.13%	RF ✓
Recall	87.76%	93.88%	RF ✓
F1-Score	87.76%	93.94%	RF ✓
CV Accuracy (5-fold)	89.75% ± 7.82%	95.49% ± 4.16%	RF ✓

6.4 Random Forest Per-Class Performance

The per-class breakdown shows how the Random Forest performs on each fire risk category. It achieves perfect recall on the Low risk class and significantly improves Severe risk detection compared to the Decision Tree.

Class	Precision	Recall	F1-Score	Support
Low	1.00	1.00	1.00	21
Moderate	0.94	0.89	0.91	18
Severe	0.82	0.90	0.86	10

6.5 Confusion Matrix Analysis

```
Predictions:
[0 0 0 0 0 1 1 1 0 0 1 1 0 0 2 0 1 1 0 1 1 2 0 1 1 0 0 0 0 1 1 2 2 2 2 2 2
 1 0 2 2 1 0 1 0 0 1 2 0]
Accuracy: 0.9387755102040817
Confusion Matrix:
[[21  0  0]
 [ 0 16  2]
 [ 0  1  9]]
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        21
           1       0.94      0.89      0.91        18
           2       0.82      0.90      0.86        10

    accuracy                           0.94        49
   macro avg       0.92      0.93      0.92        49
weighted avg       0.94      0.94      0.94        49
```

Random Forest Confusion Matrix Results:

• Low Risk: 21/21 correctly classified, achieving perfect recall.

• Moderate Risk: 16/18 correct. Only 2 misclassified as Severe.

• Severe Risk: 9/10 correct. Only 1 misclassified as Moderate, which is a major improvement over the Decision Tree which missed 3 Severe cases.

In wildfire scenarios, misclassifying Severe as Moderate (false negatives on Severe) is the most dangerous error. The Random Forest reduces this critical error from 3 (Decision Tree) to just 1, making it significantly more operationally reliable.

6.6 Feature Importance (Random Forest)

The Random Forest model provides feature importance scores, identifying the most influential variables in predicting fire ignition risk.

Rank	Feature	Importance	Category
1	ISI (Initial Spread Index)	25.76%	FWI Component
2	FFMC (Fine Fuel Moisture Code)	24.33%	FWI Component
3	DMC (Duff Moisture Code)	12.65%	FWI Component
4	BUI (Buildup Index)	12.06%	FWI Component
5	DC (Drought Code)	9.33%	FWI Component
6	Heat_Index	3.47%	Engineered
7	Rain	3.25%	Meteorological
8	Temperature	2.64%	Meteorological

Rank	Feature	Importance	Category
9	RH (Relative Humidity)	2.39%	Meteorological
10	Temp_Humidity	1.80%	Engineered

FWI system components dominate the predictions. ISI and FFMC alone account for over 50% of importance. This validates the domain-based feature engineering strategy used in the data preparation phase.

6.7 Cross-Validation Analysis

5-fold stratified cross-validation was performed on both models to assess generalization ability.

Model	CV Mean Accuracy	Std Dev	Stability Assessment
Decision Tree	89.75%	±7.82%	Moderate variance with some overfitting risk
Random Forest	95.49%	±4.16%	Low variance, stable generalizer

The Random Forest shows both higher mean accuracy (95.49%) and lower standard deviation (±4.16%) across folds compared to the Decision Tree (89.75% ±7.82%), confirming that ensemble averaging effectively reduces variance and produces a more stable generalizer.

6.8 Model Comparison & Recommendation

Criterion	Decision Tree	Random Forest
Accuracy	87.76%	93.88%
Interpretability	High (visual)	Low (black box)
Training Speed	Very Fast	Moderate
Overfitting Risk	Moderate	Low
Severe Class F1	0.70	0.86
CV Stability	Moderate	High
Recommended For	Explanation/Demo	Deployment

6.9 Key Findings & Conclusions

• Random Forest outperforms the Decision Tree across ALL evaluation metrics.

• RF achieves 93.88% accuracy vs 87.76% for DT, which is a 6.12 percentage point improvement.

• RF is significantly better at detecting Severe fires (F1: 0.86 vs 0.70), the most critical class in real-world wildfire management.

• FWI system indices (ISI, FFMC) are the most predictive features, validating the meteorological domain knowledge applied during feature engineering.

• Cross-validation confirms RF generalizes better with lower variance (±4.16% vs ±7.82%).

• Both models achieve perfect precision on the Low risk class with no dangerous false alarms for safe conditions.

Recommendation: Deploy Random Forest for operational fire risk prediction. The Decision Tree may be retained for stakeholder explanations and transparency purposes only.

Results

```
FOREST FIRE RISK PREDICTION SYSTEM
Temperature: 23
Humidity: 12
Wind Speed: 10
Rain: 12
FFMC: 11
DMC: 11
DC: 9
ISI: 19
BUI: 20
Predicted Fire Risk: Low
```

```
FOREST FIRE RISK PREDICTION SYSTEM
Temperature: 35
Humidity: 15
Wind Speed: 13
Rain: 10
FFMC: 22
DMC: 13
DC: 20
ISI: 23
BUI: 32
Predicted Fire Risk: Moderate
PS C:\Users\HP\Downloads\Forest Fire Ignition Risk>
```

```
FOREST FIRE RISK PREDICTION SYSTEM
Temperature: 40
Humidity: 15
Wind Speed: 20
Rain: 4
FFMC: 80
DMC: 49
DC: 500
ISI: 59
BUI: 80
Predicted Fire Risk: Severe
```

Chapter 7: Conclusion and Future Scope

7.1 Conclusion

This project was built around predicting forest fire ignition risk using the Algerian Forest Fire Dataset. The team worked through the full machine learning pipeline, covering data cleaning and feature engineering to model training and evaluation, using two classifiers: Decision Tree and Random Forest.

The Random Forest model came out on top with 93.88% accuracy and an F1-score of 0.86 on Severe fire risk, the most critical class. It reduced dangerous misclassifications of Severe cases from 3 to just 1 compared to the Decision Tree, which in a real wildfire situation could make a meaningful difference.

The feature importance results also showed that FWI components, especially ISI and FFMC, were the strongest predictors, accounting for over 50% of the model's decisions. This confirmed that the feature engineering done by the data team was on the right track and that meteorological indices carry real predictive value for fire risk.

7.2 Team Learnings

This project gave the whole cohort a real taste of what a machine learning workflow actually looks like. The Data Team got hands-on with messy real-world data, covering missing values, encoding categories, and building engineered features like Heat Index and Temp-Humidity Ratio that added genuine predictive value to the models.

The Modelling Team learnt how to set up and train scikit-learn classifiers, experiment with hyperparameters, and understand why ensemble methods like Random Forest tend to outperform single decision trees. The Evaluation Team went beyond just checking accuracy as they dug into confusion matrices, per-class F1-scores, and cross-validation to understand where the model was strong and where it still had gaps.

Beyond the technical work, the cohort also learnt how to function as a team remotely by splitting up responsibilities, staying in sync through daily meetings, and bringing together each sub-team's work into one clean pipeline. That kind of coordination is honestly just as important as the coding itself.

7.3 Future Scope

While the current model performs well, there is good scope to build on this work and make it more robust and practically useful in real-world settings.

1. **Address Class Imbalance:** The Severe fire risk class has only 49 samples compared to 104 Low and 90 Moderate. Techniques like SMOTE or using class_weight="balanced" during training could meaningfully improve Severe class recall.
2. **Hyperparameter Tuning:** A systematic GridSearchCV over parameters like n_estimators, max_features, and min_samples_split could push the Random Forest accuracy beyond the current 93.88%.
3. **Try Advanced Models:** Gradient boosting models like XGBoost or LightGBM are worth exploring, as they often outperform standard Random Forest on small, structured tabular datasets.
4. **Real-Time Prediction System:** The model could be wrapped into a simple web app or API that accepts live weather station inputs and returns a fire risk level instantly, making it actually deployable for forest management authorities.
5. **Add SHAP Explainability:** Adding SHAP (SHapley Additive exPlanations) analysis would let stakeholders see exactly which features drove each prediction, improving trust and transparency in operational settings.

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